

Gino Prasad

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Machine Learning | Bioinformatics | Computer Vision | Cancer Genomics | Data Science

Education

PhD in Computer Science, UC San Diego

NSF Graduate Research Fellow (Fall 2025 – Present)

Fall 2023 – Spring 2027

Academic Advisor: [Vineet Bafna](#), Professor of Computer Science, Bioinformatics and Systems Biology

GPA: 3.97 / 4.0

- **Focus:** Applied Machine Learning for Bioinformatics
- **Relevant Courses:** Deep Learning, Generative AI Modeling, Bioinformatics Algorithms, Recommender Systems, Population Genomics

Bachelor of Science, Bioinformatics: UC San Diego

Fall 2020 - Spring 2023

Major: Bioinformatics (B.S.), Minor: Computer Science

GPA 3.97 / 4.0

- **Relevant Courses:** Machine Learning, Probability & Statistics, Genomics Research Lab, Linear Algebra, Data Structures, Algorithms

Experience

Data Science & Machine Learning Intern

Jun 2025 - Aug 2025

Genentech Research & Development (gRED Computational Sciences)

- Implemented an **LLM agent system** with **LangChain** for natural language querying and analysis of large-scale cell-line screening datasets.
- Trained ML models (LightGBM, Transformer Attention Networks) to predict **drug dose-response** from gene expression and cell line embeddings.
- Utilized **SHAP** for model interpretability, gaining insights into drug potency and **mechanisms of action**.
- **Technical Skills:** LLM Agents, LangChain, Deep Learning, Transformers, PyTorch, LightGBM, SHAP

Machine Learning Graduate Researcher

Jun 2022 - Current

UC San Diego Bafna Lab

- Developed interSeg, an **AI Spatial Transcriptomics** Model for detecting ecDNA cancer presence in FISH cell imaging.
- **InterSeg** predicts cancer subtypes (ecDNA/HSR) with **single-cell resolution** in In-Vitro Cell Lines and Patient Tissue Images.
- Co-authored **EvidenceBench**, a benchmark for assessing LLM reasoning performance on biomedical papers.
- **Technical Skills:** PyTorch, Tensorflow, OpenCV, Skimage, Pandas

Computational Biology & Machine Learning Research Assistant

Oct 2021 - Jun 2023

UC San Diego Yeo Lab

- Built a **Convolutional Neural Network** for Spatial Transcriptomics Image data, to perform **nuclear segmentation**.
- Used a **U-Net Architecture** to predict nuclei boundaries, bypassing the need for DAPI staining.
- **Technical Skills:** Tensorflow, Keras, NumPy, Pandas, Pytorch, PyLab, Python, Linux, Git.

Software Engineering Intern

Jun 2021 - Aug 2021

Dotdash

- Designed front-end software for Dotdash, the largest digital publisher in the US, managing sites like Investopedia and Verywell Health.
- Developed cross-platform web applications in a collaborative environment using Agile/Scrum.
- **Technical Skills:** JavaScript, Vue, HTML, SASS, Maven, Database querying, APIs.

Phage Genomics Research Initiative

Oct 2020 - Jun 2021

UC San Diego Pogliano Lab

- Created a **BLAST parser website** using Google App Engine and Python ([GitHub](#)), used by the UCSD professor and class.
- Queries the NCBI BLAST data to perform comparative genomic analysis of Bacteriophage genes with unknown functions.
- **Technical Skills:** Flask, Python, Google Cloud App Engine.

Journal Publications

Rajkumar*, Prasad*, et al., Accurate Prediction of ecDNA in Interphase Cancer Cells using Deep Neural Networks.

bioRxiv Preprint, <https://doi.org/10.1101/2025.06.23.661188> (Co-First Author Publication)

Wang et al. (2025), EvidenceBench: A Benchmark for Extracting Evidence from Biomedical Papers.

Conference On Language Modeling, <https://doi.org/10.48550/arXiv.2504.18736>

Luebeck et al. (2025), AmpliconSuite enables discovery of extrachromosomal DNA in tumor genomes.

American Association for Cancer Research, <https://doi.org/10.1158/1538-7445.AM2025-3755>

Dehkordi et al. (2025), Breakage fusion bridge cycles drive high oncogene number with moderate intratumoural heterogeneity.
Nature Communications, <https://doi.org/10.1038/s41467-025-56670-8>

Lv et al. (2025), Spatial-Temporal Diversity of Extrachromosomal DNA Shapes Urothelial Carcinoma Evolution and Tumor-Immune Microenvironment.
Cancer Discovery, <https://doi.org/10.1158/2159-8290.CD-24-1532>

Luebeck et al. (2024), AmpliconSuite: Analyzing focal amplifications in cancer genomes.
Cancer Genetics, <https://doi.org/10.1016/j.cancergen.2024.08.015>

Mah et al. (2024), Bento: A toolkit for subcellular analysis of spatial transcriptomics data.
Genome Biology, <https://doi.org/10.1186/s13059-024-03217-7>

Chapman et al. (2023), Circular extrachromosomal DNA promotes inter- and intratumoral heterogeneity in high-risk medulloblastoma.
Nature Genetics, <https://doi.org/10.1038/s41588-023-01551-3>

Prichard et al. (2023), Identifying the core genome of the nucleus-forming bacteriophage family and characterization of Erwinia phage RAY.
Cell Reports, <https://doi.org/10.1016/j.celrep.2023.112432>

Achievements

June 2025	NSF Graduate Research Fellowship , Awarded for Outstanding Achievement in ML/Bioinformatics Research	<i>National Science Foundation (NSF)</i>
Oct 2024	Cancer Grand Challenge eDynamic Conference , Presented a Computer Vision Model for Cancer Imaging	<i>Cancer Grand Challenge (NCI)</i>
Oct 2024	Cancer Grand Challenge Future Leaders Speaker , Presented on AI applications in Cancer Research	<i>Cancer Grand Challenge (NCI)</i>
June 2023	Summa Cum Laude Honors , Awarded for Exceptional GPA.	<i>UC San Diego</i>
April 2023	Undergraduate Research Conference , Presented on Computer Vision methods for FISH Imaging.	<i>UC San Diego</i>
Jul 2020	UCSD BioScholars Honors Society Member , Awarded membership based on academic achievement.	<i>UC San Diego</i>

Skills

Machine Learning	Experience With Transformer Architectures , Convolutional Neural Networks , and ResNet Autoencoders .
Bioinformatics	Research Focus in developing AI Models for Cancer Genomics and Tumor Modeling.
Programming	Python (PyTorch, Tensorflow, Keras, Pandas, NumPy), R, C++, Cloud Computing, Bash, Linux, Git, JavaScript, Java, SQL.

Mentoring

2023 - 2024	Data Science Capstone Mentor , Mentored 3 Data Science undergrads in a Computer Vision capstone competition to semantically segment cell imaging data using deep learning.	<i>UC San Diego</i>
2023 - 2024	Early Research Scholars Program Mentor , Mentored 2 Computer Science undergrads in a project to extract mutational signatures from The Cancer Genome Atlas (TCGA) genome sequencing data.	<i>UC San Diego</i>

Personal Projects

Autotune Implementation Using Phase Vocoder  github.com/GinoP123/AutotunePV.git - Created an autotuner from scratch using Phase Vocoders and Yin pitch prediction. - Able to autotune any audio clip to a specific major or minor scale using Hann window functions. - Examples of popular songs autotuned here.	<i>May 2023</i>
Custom Search Engine for Linux File System  github.com/GinoP123/FileSearch - Created a keyword-matching search engine with caching fully from scratch using dynamic programming. - Added learning capability by including popularity and relevance weights. - I personally use this tool all the time, and find it a huge time-saver for navigating in Linux.	<i>Jul 2022</i>